Take-Home Assignment:

The aim of this assignment is to evaluate your acumen and aptitude towards building and owning the analytical & decision framework for aircraft assignment related problems which might arise to solve them using Reinforcement Learning.

Time box: Fewer than 6 hours, we are not interested in you spending enumerable hours in building this but would like to evaluate your reasoning, judgements, and knowledge

Scenario: We dispatch a fleet of battery-electric aircraft across a city network. Unlike road transport, every decision is coupled through hard energy limits, charging time and scarce pad capacity. We want to know whether learned policies can improve on hand-written dispatch heuristics. If they can, we want to know when they can and how they fail. If they can't, we want to know that too.

Further details of the scenario are noted in the Scenario.docx file attached to this. Your problem is now 2-fold:

Problem A: Build a simulation environment for the network that implements the scenario.docx mentioned above

Minimum requirements for the simulator are:

1. Energy, reserve, charging curve, pad capacity, time windows and the 30-minute booking lead time are all enforced.
2. The environment can replay the fixed 70-request instance and can also sample new days from the demand generator.
3. Deterministic Seeding: the same seed and the same actions produce the same trajectory.
4. An independent constraint checker that takes a completed episode log and verifies every rule. Do not just reuse the environment's own internal checks

Problem B: Train a policy for this problem.

- Any method is fine: PPO, DQN, imitation learning from Greedy-JIT followed by fine-tuning, and so on.
- Training must complete in ≤ 45 minutes on a laptop CPU. Tell us your hardware.
- Safety constraint: the deployed policy must never violate the energy reserve or accept a request it then fails. Decide whether this is enforced by action masking, a shield or a penalty. Explain why, including what goes wrong with the alternatives.
- Document the reward design. Include at least one reward variant you tried or considered and rejected, and why

Compare your policy against JIT. In addition, also focus on sensitivity to demand data and report on:

a) Impact on Profit, Requests served, failures, deadhead km, no of battery charge-discharge cycles
b) Inference Time for each run through
c) Failure cases

Write a short note covering:

a) Your modeling choice, the framework and your evaluation approach
b) The comparison and evaluation framework and their utilisation
c) Sensitivity analysis of the model to small perturbations to the input

Please use the provided jit.py as the implementation for the JIT and use it for comparison and evaluation as and when needed

GenAI tool use

Use whatever tools you'd normally use, including LLMs. We're not going to pretend you'd work without GenAI on the job. What we care about is that you understand and can defend every decision in your submission

Please include a short note stating what AI tools you used and roughly what for. Honest disclosure here costs you nothing. What will cost you is being unable to explain or extend your own submission in the follow-up conversation.

Expected Output:

1. Raw outputs and results
2. Code repository with trained policy weights, training scripts with seeds, and environment implementation
3. The short note as mentioned above
4. AI-tool disclosure

Assessment Framework:

Area	What we look for
Environment & correctness	Rules implemented faithfully; Independent checker; Determinism;

RL design	Sensible state, action and reward design; safety enforced by construction
Evaluation rigour	Held-out data, multiple seeds, distribution shift, honest comparison to baseline
Failure-mode understanding	Concrete failures found and explained; clear statement of where this would be unsafe to deploy
Engineering quality	Readable, runnable, reproducible; sensible tests